

Repurposing Approved Drugs for Polycystic Ovary Syndrome by Network Proximity Analysis of Large-Scale Drug-Target and Protein-Interaction Data

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Abstract

Polycystic ovary syndrome (PCOS) affects about one in ten women of reproductive age, yet no drug has been approved specifically for it. Current therapy borrows drugs from diabetes, hypertension and infertility, each identified by clinical intuition rather than systematic search. This proposal describes a computational pipeline that ranks all approximately 2,800 approved small-molecule drugs by the network proximity of their protein targets to the PCOS disease module in the human interactome. The pipeline integrates ChEMBL bioactivity data (20 million measurements), Open Targets and GWAS Catalog disease evidence, and the STRING protein-interaction network (12 million links). The proximity computation, about 2.8 million shortest-path evaluations with degree-matched randomisation, exceeds single-machine memory and is implemented as a distributed Apache Spark job. Drugs already used for PCOS are held out and used to validate the ranking. The deliverable is an evidence-annotated candidate list with a stated numeric success criterion.

Keywords: drug repurposing, PCOS, network medicine, Apache Spark, ChEMBL, Open Targets, STRING.

1. Introduction and Biological Problem

Polycystic ovary syndrome is the most common endocrine disorder in women of reproductive age, with a pooled prevalence of roughly 10% (Bozdag et al., 2016). It presents through three interacting biological axes: insulin resistance, hyperandrogenism and chronic low-grade inflammation (Escobar-Morreale, 2018). Despite this burden, no drug carries a regulatory approval for PCOS. The 2023 international guideline recommends metformin (a diabetes drug), spironolactone (an antihypertensive), combined oral contraceptives, and letrozole or clomiphene for ovulation induction (Teede et al., 2023). All of these are borrowed, symptomatic treatments.

The gap. Each borrowed drug was identified individually, from clinical observation. There has been no systematic, evidence-weighted screen of the entire approved pharmacopoeia against PCOS biology. Because de novo drug development takes 10 to 15 years and costs over one billion US dollars (Di Masi et al., 2016), while an approved drug already has human safety and pharmacokinetic data

(Pushpakom et al., 2018), a data-driven repurposing screen is the fastest route to new PCOS treatment options.

Objective. To produce a ranked, evidence-annotated list of approved drugs whose protein targets are significantly closer than chance to the PCOS gene module in the human protein-interaction network, and to validate that ranking against drugs already in PCOS use.

2. Why This Is a Big Data Problem

The PCOS gene set itself is modest, about 20 genome-wide significant loci (Day et al., 2018) and a few hundred Open Targets associations. The scale of this project arises from the drug side and the network side, and can be stated in terms of the four Vs.

- **Volume.** ChEMBL contains 20 million bioactivity measurements over 2.4 million compounds and 15,000 targets (about 25 GB loaded). STRING human contains 12 million scored protein links. Open Targets distributes 18 million evidence strings, about 50 GB in Parquet.
- **Variety.** Relational bioactivity tables, disease-evidence JSON, a weighted graph, drug-annotation XML and clinical-trial records, each using a different identifier system (ChEMBL, UniProt, Ensembl, STRING, EFO).
- **Veracity.** Experimental versus predicted interactions, evidence scores on a 0 to 1 scale, and sources with restrictive licences (KEGG, DisGeNET bulk) that must be excluded and substituted.
- **Velocity.** ChEMBL and Open Targets are released quarterly, so the pipeline must be reproducible and re-runnable rather than a one-off analysis.

The step that exceeds a single machine. Network proximity must be computed for each of about 2,800 drugs against a 19,600-node, 12-million-edge graph, and each drug's score must be compared with 1,000 degree-matched random target sets. This amounts to roughly 2.8 million shortest-path computations. The graph plus per-drug state exceeds 8 GB of RAM, and a single core would need days. The problem is embarrassingly parallel across drugs, which maps naturally to a Spark *map* over drug partitions with the graph as a broadcast variable. Spark is preferred to Hadoop MapReduce because the computation is iterative and Spark retains intermediate state in memory rather than writing to disk each round (Zaharia et al., 2016; Dean & Ghemawat, 2008). Open Targets already ships Parquet, Spark's native columnar format.

3. Data Sources

Source	Role	Size and format	Example record	Licence
ChEMBL v35 (Zdrazil et al., 2023)	Drug-target bioactivities	20 M activities, 15 k targets; ~4 GB compressed, Parquet	CHEMBL25 (aspirin) → PTGS1, IC50 = 32 nM, pchembl 7.5	CC BY-SA
DrugBank (Knox et al., 2023)	Approval status, indications	16.5 k drugs, 2.8 k approved; XML 1.5 GB	DB00331 metformin, approved, target PRKAB1, indication T2D	Academic
Open Targets Platform (Buniello et al., 2024)	PCOS-gene associations, evidence scores	18 M evidence strings; associations ~2 GB Parquet	ENSG00000105329 (TGFB1), EF0_0000660, score 0.42	CC0
GWAS Catalog (Sollis et al., 2022)	Genetic seed genes	700 k associations; TSV ~300 MB	rs11031006, FSHB, PCOS, p = 1×10^{-12}	Open
STRING v12, human (Szklarczyk et al., 2022)	Protein-interaction network	19.6 k proteins, 12 M links; TSV	9606.ENSP00000284981 – 9606.ENSP00000262607, score 912	CC BY 4.0
UniProt / Ensembl (The UniProt Consortium, 2025)	Identifier mapping	20 k human proteins; mapping files, hundreds of MB	P37231 ↔ ENSG00000132170 ↔ PPARG	CC BY 4.0
ClinicalTrials.gov (NLM)	Validation set	500 k trials; JSON/CSV	NCT id, metformin, condition PCOS, phase 3	Open

Table 1. Public data sources, their role in the pipeline, approximate size, one example record and licence.

Two substitutions were made after a licence check. KEGG bulk download is licensed, so Reactome is used for pathway context. DisGeNET now restricts bulk access, so the disease side is built on Open Targets and DisGeNET is cited only as a cross-check.

4. Methodology

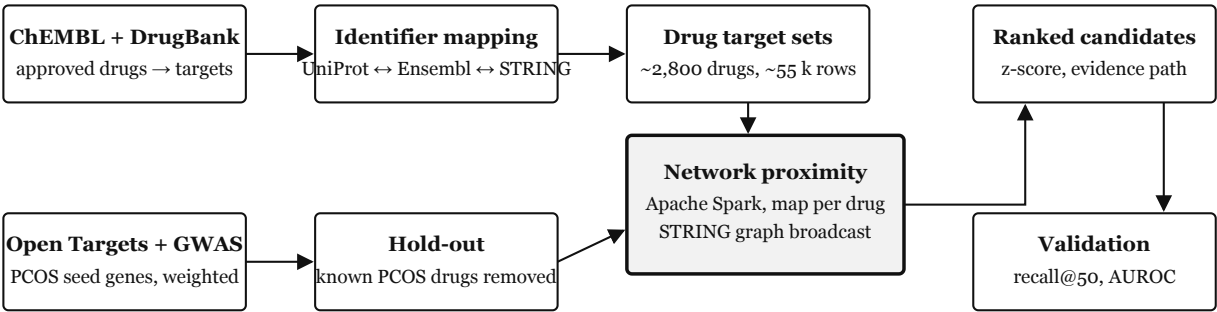


Figure 1. Pipeline overview. The shaded step is the distributed computation; all other steps produce small outputs.

- Ingestion.** Download ChEMBL, DrugBank, Open Targets associations, GWAS Catalog and STRING; store all tables as Parquet in HDFS or local object storage.
- Approved drug set.** Select ChEMBL molecules with `max_phase = 4`, cross-checked against the DrugBank "approved" flag. Expected output: about 2,800 drugs.
- Drug target sets.** Spark join over 20 million ChEMBL activity rows, retaining human single-protein targets with `pchembl ≥ 6` (potency $\leq 1 \mu\text{M}$). Output: each drug mapped to a set of UniProt accessions.

4. **PCOS seed set.** Open Targets associations for EF0_0000660 with overall score ≥ 0.1 , plus GWAS Catalog mapped genes, each weighted by its evidence score. Expected 300 to 500 genes.
5. **Identifier mapping.** Map UniProt to Ensembl to STRING protein identifiers. Every unmapped identifier is logged and the loss rate reported.
6. **Hold-out.** Before scoring, drugs already used or in trials for PCOS (metformin, spironolactone, letrozole, clomiphene, pioglitazone, liraglutide, statins) are removed from seed construction and reserved as the validation set.
7. **Network proximity.** Following Guney et al. (2016), compute for each drug the mean shortest-path distance from its targets to the seed set on STRING (combined score ≥ 400), and compare it with 1,000 degree-matched random target sets to obtain a z-score. Implemented as a Spark map over drug partitions with the graph broadcast to each executor.
8. **Ranking and annotation.** Sort drugs by z-score and attach current indication, the closest PCOS genes and the evidence type supporting each link.
9. **Validation.** Measure recall of the held-out PCOS drugs in the top 50 and the AUROC against all other approved drugs, following the trial-based validation of Cheng et al. (2018).

5. Expected Outcome

Deliverable. A ranked table of approved drugs with proximity z-score, current indication, nearest PCOS genes and evidence type, together with a network figure of the top ten candidates and their paths into the PCOS module.

Success criterion. At least four of the approximately eight held-out PCOS drugs should appear in the top 50 of about 2,800 (fewer than 0.2 expected by chance), with AUROC above 0.7. Should the hold-out not be recovered, the negative result still establishes whether PCOS biology is reachable through the currently druggable target space, which is itself a useful finding.

6. Challenges and Limitations

- **Identifier mapping** across five sources is the hardest engineering step. A 5 to 10% loss of targets is typical and will be measured and reported rather than hidden.
- **Small, noisy disease gene set.** Seeds are weighted by evidence type, and an ablation tests whether genetic seeds alone recover the hold-out better than all seeds combined.
- **Literature bias.** Well-studied proteins have more edges and more evidence, so raw proximity favours them. Degree-matched randomisation controls for this.
- **Data leakage.** If known PCOS drugs remained in seed construction, validation would be circular. Holding them out before scoring removes this risk.
- **Limitation.** The output is a computational prioritisation. Any candidate requires experimental and clinical validation before it has any bearing on patient care.

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